THE UNITED REPUBLIC OF TANZANIA
MINISTRY OF DEFENCE AND NATIONAL SERVICE
MILITARY SCHOOLS ASSOCIATION
FORM TWO MOCK ASSESSMENT

MARKING SCHEME

COMPUTER SCIENCE

SECTION A (15 Marks)

Answer all questions in this section.

QUESTION 1 (10 Marks @ 1 mark each)

Each multiple-choice question carries 1 mark.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	B	B	B	C	A	A	B	B	B

QUESTION 2 (5 Marks)

Each correct match carries 5/6 of a mark (6 items sharing 5 marks).

(i)	(ii)	(iii)	(iv)	(v)	(vi)
E	C	B	F	D	A

SECTION B (70 Marks)

Answer all questions in this section.

QUESTION 3 (10 Marks)

(a) Why first generation computers were large, over-heating, and power inefficient. (2 marks)

Answer: First generation computers used vacuum tubes as their main electronic component. Vacuum tubes are physically bulky, generate a great deal of heat during operation, and consume large amounts of electrical power, which together made these computers large in size, prone to overheating, and inefficient in power use.

Marking:

- o 2 marks: Correctly links the cause (vacuum tube technology) to size, heat, and power inefficiency.
- o 1 mark: Mentions vacuum tubes or one consequence only, without full linkage.

(b) Why the development of microprocessors was a major development in computer technology. (2 marks)

Answer: The microprocessor integrated the entire processing unit (control unit and ALU) onto a single small chip. This made computers much smaller, cheaper, faster, and more power-efficient, and it made possible the mass production of personal computers and other digital devices.

- o 2 marks: Explains integration onto a single chip and the resulting benefits (size, cost, speed).
- o 1 mark: States it made computers smaller/cheaper without explaining why.

(c) Why a mobile phone is regarded as a computer. (Three reasons) (3 marks)

Answer: A mobile phone is regarded as a computer because:

- It has a processor (CPU) that processes instructions.
- It has memory and storage for holding data and programs.
- It runs an operating system and can install and run application software (apps).
- It accepts input (touchscreen, microphone, camera) and produces output (screen, speaker).

o 1 mark for each valid, well-stated reason, up to a maximum of 3 marks.

(d) Why most people prefer a laptop over a desktop computer. (Three reasons) (3 marks)

- Portability: a laptop is light and compact, so it can be carried and used anywhere.
- Built-in battery: allows it to be used during a power outage or while travelling.
- All-in-one design: keyboard, screen, camera and speakers are built in, saving space and cabling.
- Convenience for work and study away from a fixed desk.

o 1 mark for each valid, well-stated reason, up to a maximum of 3 marks.

QUESTION 4 (10 Marks)

(a) Why the teacher opted to use To, CC and BCC fields. (Three reasons) (3 marks)

- To clearly separate the main recipient(s) who must act (To) from those who are only being informed (CC).
- To keep a transparent record of everyone involved in the communication.
- To protect the privacy of some recipients by hiding their addresses from others (BCC).
- To reach many students at once efficiently through a single email.

o 1 mark for each valid reason, up to a maximum of 3 marks.

(b) Function of each field appearing when composing an e-mail. (3 marks)

- To field: Contains the address(es) of the main/primary recipient(s) who are expected to read and act on the message.
- CC (Carbon Copy) field: Contains addresses of people who should be kept informed of the message; all recipients can see who is in the CC list.
- BCC (Blind Carbon Copy) field: Contains addresses of recipients who receive a copy without other recipients knowing they were included.

o 1 mark for each correctly explained field.

(c) Why it is important to use CC and BCC fields appropriately when sending emails. (4 marks)

- Protects the privacy of recipients' email addresses (avoids exposing everyone's address to strangers).
- Prevents unnecessary clutter/spam from mass 'reply-all' responses.
- Maintains professionalism and confidentiality in official communication.
- Avoids misuse of email addresses for unsolicited messages or malicious purposes.

o 1 mark for each valid, well-explained point, up to a maximum of 4 marks.

QUESTION 5 (10 Marks)

(a) Name the parts indicated by letters A to H. (4 marks)

Answer: Based on the standard block diagram of a computer system:

- A – System/Central Processing Unit (CPU) boundary (the processor as a whole)
- B – Control Unit (CU)
- C – Arithmetic and Logic Unit (ALU)
- D – Registers / Internal storage of the CPU
- E – Main (Primary) Memory / Storage Unit
- F – System Bus (control/data/address pathway)
- G – Input Unit
- H – Output Unit

o 0.5 mark for each correctly named part (8 parts), maximum 4 marks.

(b) Functions of the parts labelled B and C. (3 marks)

- B – Control Unit: Directs and coordinates the activities of the computer; it fetches, decodes and directs the execution of instructions, and controls the flow of data between the CPU, memory and input/output devices.
- C – Arithmetic and Logic Unit (ALU): Performs all arithmetic operations (addition, subtraction, etc.) and logical operations (comparisons such as AND, OR, greater-than/less-than).

o 1.5 marks for each correctly explained function.

(c) Steps to install the CPU/processor (part A) on the motherboard of a desktop computer. (3 marks)

• Switch off and unplug the computer, and ground yourself (use an anti-static wrist strap) to avoid static damage.

• Open the computer case and locate the CPU socket on the motherboard.

• Release/lift the socket retention lever or lock.

• Align the processor correctly using its notch/triangle marker with the corresponding mark on the socket.

• Gently lower the processor into the socket without forcing it, then close and secure the retention lever.

• Apply a small amount of thermal paste on top of the processor.

• Mount the heatsink/fan on top of the processor and secure it, then connect the fan's power cable to the motherboard.

o 3 marks: Steps given in a logical, complete order.

o 2 marks: Most steps correct but one omission/out of order.

o 1 mark: A basic attempt with major gaps.

QUESTION 6 (10 Marks)

(a) Advantages and disadvantages of SSDs compared to HDDs. (5 marks)

Advantages of SSD over HDD:

• Much faster read/write (data access) speed.

• No moving parts, so more durable/shock-resistant.

• Silent operation (no spinning disk noise).

• Lower power consumption and less heat generated.

• Lighter and smaller in size.

Disadvantages of SSD compared to HDD:

• More expensive per gigabyte of storage.

• Generally offers lower maximum storage capacity for the same price.

• Limited number of write/erase cycles over its lifespan.

o 1 mark for each valid, correctly stated advantage or disadvantage, maximum 5 marks.

(b) Five factors Dickson should consider before installing software. (5 marks)

• System requirements/compatibility: whether the computer's hardware and operating system meet the software's minimum requirements.

• Available storage space on the disk for the installation.

• Source/legitimacy of the software: obtaining it from a trusted, licensed source to avoid malware.

• Compatibility with existing software (no conflicts with programs already installed).

• Administrative rights/permissions needed to install the software.

• Backing up important data before installation, in case something goes wrong.

o 1 mark for each valid factor, maximum 5 marks.

QUESTION 7 (10 Marks)

(a) Define 'computer handling and care' and give three points on its importance. (5 marks)

Answer: Computer handling and care refers to the proper practices, precautions and habits followed when using, cleaning, moving and storing a computer, aimed at protecting it from damage and keeping it working efficiently.

Importance:

• Prevents physical damage to hardware components, prolonging the computer's lifespan.

• Protects the safety of the user (avoids electric shock, injury, eye strain).

• Maintains good performance and reliability of the system.

• Protects data stored on the computer from loss or corruption.

o 2 marks for a clear, complete definition (1 mark for a partial definition).

o 1 mark for each valid importance point, maximum 3 marks.

(b) Four functions of Uninterruptible Power Supplies (UPS). (5 marks)

• Provides temporary backup power to connected devices during a power outage.

• Protects equipment from power surges, spikes and voltage fluctuations.

• Gives the user enough time to save work and shut down the computer safely when power fails.

• Stabilises/regulates the incoming power supply to the computer.

o 1.25 marks for each correctly stated function, maximum 5 marks.

QUESTION 8 (10 Marks)

(a) A printer is not responding to print requests. (2.5 marks)

• Check that the printer is powered on and all cables/wireless connection are properly connected.

• Check that the correct printer is selected as the default printer.

• Clear the print queue of stuck jobs and resend the print job.

• Check for paper jams or low ink/toner, and restart the printer and computer if necessary.

(b) An external hard drive is not showing up on the computer. (2.5 marks)

• Check that the USB cable and port are properly connected; try a different cable or port.

• Check Disk Management (or equivalent) to see if the computer detects the drive but it lacks a drive letter/is not initialised.

• Ensure the drive has adequate power (some external drives need external power).

• Test the drive on another computer, and update/reinstall the disk drivers if needed.

(c) A computer monitor displays a distorted or no image at all. (2.5 marks)

• Check that the video cable is firmly connected at both the monitor and the computer.

• Check the monitor's power connection and that it is switched on.

• Check and adjust the display resolution/refresh rate settings.

• Test with another monitor or cable to isolate the fault, and check that the graphics card is properly seated.

(d) A student's computer runs slower than usual and cannot save work. (2.5 marks)

• Check available disk space and delete unnecessary files to free up storage.

• Scan the computer for viruses/malware using antivirus software.

• Close unnecessary running programs/background processes using Task Manager.

• Run disk cleanup/defragmentation and restart the computer.

Marking:

o 2.5 marks per part: award marks for each correct, logically ordered troubleshooting step given (up to the mark allocation shown), accepting any other valid diagnostic step.

QUESTION 9 (10 Marks)

(a) Three possible ways the account might have been hacked via public Wi-Fi. (6 marks)

• Man-in-the-Middle attack: the attacker intercepts data (including login details) being transmitted between the user's device and the network.

• Fake/rogue hotspot ('evil twin'): the attacker sets up a Wi-Fi network with a legitimate-looking name to trick users into connecting, then captures their data.

• Packet sniffing: the attacker uses software to capture unencrypted data travelling over the open network, including usernames and passwords.

• Malware distribution: the attacker uses the unsecured network to push malicious software onto the connected device.

o 2 marks for each correctly explained way, maximum 6 marks.

(b) Four measures to protect a social media account when using public internet. (4 marks)

- Use a Virtual Private Network (VPN) to encrypt internet traffic.
- Avoid logging into sensitive accounts or entering personal/financial information on public Wi-Fi.
- Ensure websites use HTTPS (secure connection) before entering any credentials.
- Enable two-factor authentication (2FA) on the account, and log out and forget the network after use.

o 1 mark for each valid measure, maximum 4 marks.

SECTION C (15 Marks)

Answer question ten (10).

QUESTION 10 (15 Marks)

In four points, explain how the computer system confirms that Kole is the authorised person to access and perform transactions.

Answer: The system may confirm Kole's identity through:

- Username and password verification: Kole enters a unique username with a secret password that only he should know; the system checks these against stored records before granting access.
- Two-factor/multi-factor authentication (2FA/MFA): after the password, the system sends a one-time code (OTP) to Kole's registered phone number or email, which he must also enter to confirm it is really him.
- Biometric authentication: the system may verify Kole's fingerprint, face, or other biometric feature captured by the mobile phone before allowing access to the banking app.
- PIN verification: Kole enters a personal identification number (PIN) known only to him to authorise the specific payment transaction.
- Digital certificates/encryption: the transaction is secured and the identity of the device/session is verified using encryption keys or digital certificates issued to Kole's account.

Marking:

o 3.75 marks for each well-explained point (up to 4 points required), Total 15 marks.

o Award partial marks for a point that is identified but not well explained.

o Accept any other plausible, well-argued authentication method (e.g. security questions, session timeout/lockout after failed attempts).